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| **Status** | Finished |
| **Started** | Sunday, 18 October 2026, 12:37 AM |
| **Completed** | Sunday, 18 October 2026, 12:41 AM |
| **Duration** | 3 mins 55 secs |
| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By using the bubble-up method
- c. By breaking the input into smaller parts and merging them
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. All of the above
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Sorting

c. Complexity

d. Searching

The correct answer is:
Complexity

Question **5**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Bubble sort

c. Heap sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **6**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection

b. Extraction

c. Insertion

d. Distribution

The correct answer is:
Insertion

Question **7**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items

b. It makes the data unsorted

c. It slows down the process

d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Heap Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Merging two datasets
- c. Arranging data in a specific order
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming

b. Greedy method

c. Backtracking

d. Divide and conquer

The correct answer is:
Divide and conquer

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation

b. It makes the search operation slower

c. It increases the number of comparisons needed

d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Sorting the entire list sequentially
- c. Comparing each element with the others
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It divides the list into two halves, sorts each half, and then merges them
- c. It compares adjacent elements and swaps them if necessary
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Inserting data into a list
- c. Searching for data in a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format